JUNIOR SCHOOL LEARNING HUB

GRADE 9 AGRICULTURE
END-OF-TERM 3 EXAMINATION

Name: _____	Adm. No.: _____	Date: _____	Class: _____
Time: 2 hours 30 minutes	Total: 100 marks	Term: 3	Learning Area: Agriculture

INSTRUCTIONS TO LEARNERS

- Answer all questions.
- Read each question carefully.
- Where a diagram is required, draw it clearly and label it correctly.
- Show all working where calculations are required.

SECTION A: SHORT ANSWER QUESTIONS (40 MARKS)

1. Define agroforestry. (2 marks)

2. State two benefits of planting trees on a farm. (2 marks)

3. Give two examples of indigenous trees that may be useful on Kenyan farms. (2 marks)

4. State two factors to consider when selecting a site for a tree nursery. (2 marks)

5. Differentiate between a seedbed and a potting container. (2 marks)

6. Name two materials used in preparing a suitable potting mixture. (2 marks)

7. State two ways of protecting young tree seedlings from strong sunlight or wind. (2 marks)

8. What is transplanting in tree establishment? (2 marks)

9. Give two reasons for hardening off seedlings before transplanting. (2 marks)

10. State two precautions to observe when pruning trees. (2 marks)

11. Define soil and water conservation. (2 marks)

12. Name two physical structures used to control soil erosion on a farm. (2 marks)

13. State two farming practices that help reduce soil erosion. (2 marks)

14. Give two causes of water pollution on farms. (2 marks)

15. State two ways of conserving water for agricultural use. (2 marks)

16. What is rainwater harvesting? (2 marks)

17. Name two methods of applying irrigation water to crops. (2 marks)

18. State two signs of water stress in a crop. (2 marks)

19. Give two advantages of mulching. (2 marks)

20. State two safe practices when handling agricultural chemicals. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

21. Tree nursery management (12 marks)

(a) Explain four factors to consider when choosing a site for a tree nursery. (4)

(b) Describe four steps followed when preparing a potting mixture and filling containers. (4)

(c) Explain two management practices that promote healthy seedling growth. (4)

22. Establishment and management of trees (12 marks)

(a) Outline four steps followed when transplanting a tree seedling to the field. (4)

(b) Explain four practices that should be carried out after transplanting. (4)

(c) State four benefits of agroforestry to a farming household. (4)

23. Soil erosion and conservation (12 marks)

(a) Describe four ways in which soil erosion reduces agricultural productivity. (4)

(b) Explain four methods farmers can use to control soil erosion. (4)

(c) State four indicators that may show that soil conservation measures are working. (4)

24. Water conservation and irrigation (12 marks)

(a) Explain four methods of harvesting or conserving water on a farm. (4)

(b) Differentiate between drip irrigation and sprinkler irrigation. (4)

(c) Give four factors to consider when choosing an irrigation method. (4)

25. Sustainable and safe farm practices (12 marks)

(a) Explain four ways of preventing pollution of water sources on a farm. (4)

(b) Describe four precautions to take when storing or applying agricultural chemicals. (4)

(c) Explain four ways in which mulching supports sustainable crop production. (4)

MARKING SCHEME

Grade 9 Agriculture – End-of-Term 3 Examination

Total: 100 marks | Section A: 40 | Section B: 60

SECTION A: SHORT ANSWER QUESTIONS (40 MARKS)

1. A farming system that deliberately combines trees or shrubs with crops and/or livestock on the same land. (2)

2. Any two: reduce wind speed; provide shade; protect soil; supply fruits, fodder, fuelwood or timber; improve biodiversity. (1 each)

3. Any two suitable examples, e.g. Croton megalocarpus, Markhamia lutea, Warburgia ugandensis, Prunus africana. (1 each)

4. Any two: reliable water supply; well-drained soil; accessible location; suitable sunlight/shade; protection from animals and strong winds. (1 each)

5. A seedbed is a prepared area where seeds are sown and germinated; a potting container is an individual bag, pot or tube holding growing medium for a seedling. (1 each)

6. Any two: topsoil, well-decomposed compost/manure, sand, or other locally suitable materials. (1 each)

7. Any two: temporary shade; windbreaks; shade netting; careful positioning of containers. (1 each)

8. Moving a young seedling from a seedbed/container to its permanent growing site. (2)

9. Any two: helps seedlings adapt gradually to outdoor conditions; reduces transplant shock; improves survival. (1 each)

10. Any two: use clean, sharp tools; remove only necessary branches; avoid damaging the trunk; prune at suitable times; use safe handling practices. (1 each)

11. Practices that prevent loss of soil and protect water resources from depletion or contamination. (2)

12. Any two: terraces, contour bunds, cut-off drains, grassed waterways, check dams. (1 each)

13. Any two: contour farming; cover cropping; mulching; strip cropping; minimum tillage; planting grass strips. (1 each)

14. Any two: pesticide/fertiliser runoff; animal waste; oil or fuel spills; dumping chemicals or refuse; erosion sediments. (1 each)

15. Any two: roof catchment and storage; farm ponds; water tanks; mulching; efficient irrigation; repairing leaks. (1 each)

16. Collecting and storing rainwater from surfaces such as roofs or paved catchments for later use. (2)

17. Any two: drip, sprinkler, surface/furrow, basin irrigation. (1 each)

18. Any two: wilting; leaf rolling; yellowing; stunted growth; premature leaf fall. (1 each)

19. Any two: conserves soil moisture; suppresses weeds; reduces erosion; moderates soil temperature; adds organic matter when decomposed. (1 each)

20. Any two: wear recommended protective equipment; follow label instructions; avoid eating or drinking; keep chemicals away from children; wash after handling; dispose of containers safely. (1 each)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

21. Tree nursery management (12 marks)

(a) Any four, 1 mark each: accessible to workers; reliable water; well-drained site; suitable light/shade; protected from livestock; near planting area/road; not prone to flooding. (4)

(b) Any four steps, 1 mark each: obtain suitable clean materials; remove stones/debris; mix topsoil with mature compost/manure and suitable sand; moisten lightly; fill containers without excessive compaction; arrange containers upright. (4)

(c) Any two practices explained, 2 marks each: regular watering without waterlogging; weeding; pest/disease monitoring; appropriate shading; spacing; removal of weak seedlings. (4)

22. Establishment and management of trees (12 marks)

(a) Any four, 1 mark each: water seedling before moving; prepare planting hole; remove container carefully; place seedling upright at correct depth; refill with soil; firm gently; water after planting. (4)

(b) Any four, 1 mark each: water regularly; mulch while keeping mulch away from stem; replace dead seedlings; weed; protect from livestock; stake if needed; inspect for pests and diseases. (4)

(c) Any four, 1 mark each: provides fruit/fodder/fuelwood/timber; improves soil protection; supplies shade/windbreaks; supports biodiversity; diversifies household income; improves microclimate. (4)

23. Soil erosion and conservation (12 marks)

(a) Any four, 1 mark each: removes fertile topsoil; carries away nutrients; exposes roots; forms gullies; reduces water infiltration; deposits sediment in waterways; lowers yields. (4)

(b) Any four methods with a brief explanation, 1 mark each: terracing; contour ploughing; cover crops; mulching; grass strips; agroforestry; controlled grazing; minimum tillage. (4)

(c) Any four, 1 mark each: less rill/gully formation; reduced sediment in drains; better soil moisture; improved crop growth; stable topsoil; increased vegetation cover. (4)

24. Water conservation and irrigation (12 marks)

(a) Any four, 1 mark each: roof catchment into tanks; farm ponds; lined water pans; contour bunds; mulching; repair leaks; reuse suitable greywater safely; protect catchments. (4)

(b) Drip irrigation delivers water slowly near plant roots through emitters; sprinkler irrigation sprays water over crops through pressurised nozzles. Award 2 marks for each accurate description. (4)

(c) Any four, 1 mark each: water availability; soil type; crop type; slope/topography; cost; labour and skills; energy needs; water quality; maintenance. (4)

25. Sustainable and safe farm practices (12 marks)

(a) Any four, 1 mark each: apply chemicals at recommended rates; avoid spraying near water; maintain buffer strips; store manure away from streams; prevent soil erosion; dispose of waste correctly; maintain equipment to prevent leaks. (4)

(b) Any four, 1 mark each: retain labelled original containers; lock storage away from children and animals; keep away from food and water; wear PPE; follow label directions; avoid spraying in strong wind; wash equipment safely; dispose of containers as directed. (4)

(c) Any four, 1 mark each: reduces evaporation; suppresses weeds; reduces splash erosion; moderates soil temperature; improves soil structure as organic mulch decomposes; supports soil organisms. (4)

ADMINISTRATION NOTE

This is an original teacher-prepared assessment resource for practice and school-based assessment. Before administration, the Agriculture teacher should confirm that the topics assessed match the actual Grade 9 Term 3 teaching sequence and the school scheme of work. Accept other scientifically and agriculturally correct responses.

Prepared by: Junior School Learning Hub